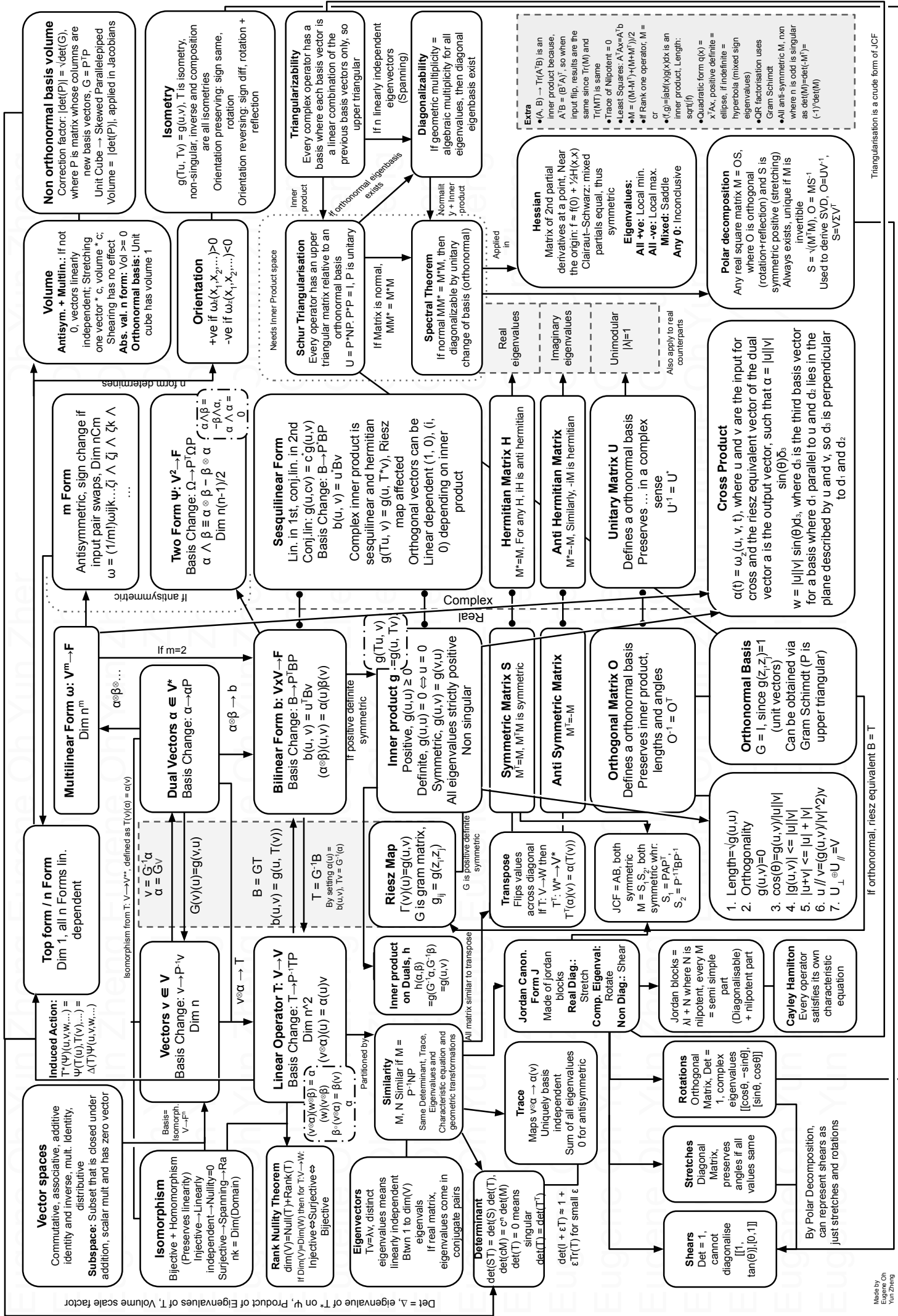




Fact 1.0: vector element of space $\mathbb{R}^2/\mathbb{R}^3$ with magnitude direction, addable scalable
Fact 1.1: linear transformation preserves structure $T(u+v)=T(u)+T(v)$, $T(cu)=cT(u)$, maps lines→lines
Fact 1.2: identity $I(u)=u$
Fact 1.3: scaling multiplies all vectors by scalar
Fact 1.4: stretching along u scales parallel component, leaves perpendicular unchanged
Fact 1.5: rotation preserves length/angle changes direction
Fact 1.6: linear transformation determined by action on basis
Fact 1.7: matrix columns = images of basis vectors
Fact 1.8: composition = matrix product $AB = B$ then A, generally $AB \neq BA$
Fact 1.9: $\det 2D = \text{area scale}$, $\det 3D = \text{volume scale}$
Fact 1.10: $\det 0 \Leftrightarrow \text{collapse dimension} \Leftrightarrow \text{singular} \Leftrightarrow \text{not invertible}$
Fact 1.11: non-singular $\Leftrightarrow \det \neq 0 \Leftrightarrow \text{invertible}$
Fact 1.12: inverse reverses map $T^{-1}T=I$, $(AB)^{-1}=B^{-1}A^{-1}$
Fact 1.13: linear system $Mx=a$ matrix form
Fact 1.14: $\det M \neq 0 \Leftrightarrow \text{unique solution}$, $\det M=0 \Leftrightarrow \text{none or infinite solutions}$
Fact 1.15: eigenvector $Tu=\lambda u$ direction unchanged, eigenvalue λ scale factor
Fact 1.16: eigenvalues from $\det(T-\lambda I)=0$, eigenvectors from nullspace
Fact 1.17: eigenvectors not unique (any scalar multiple valid)
Fact 1.18: diagonalizable $\Leftrightarrow$ eigenvector basis exists $\Leftrightarrow$ matrix becomes diagonal
Fact 1.19: non-diagonalizable cases exist, eigenbasis may fail
Fact 1.20: trace = sum diagonal entries = sum eigenvalues
Fact 1.21: determinant = product eigenvalues, basis-independent
Fact 1.22: change of basis: $P^{-1}AP$
Fact 1.23: basis = independent spanning vectors
Fact 1.24: coordinate change uses P and P^{-1}
Fact 1.25: 2×2 inverse = $1/(ad-bc)[d \ -b; -c \ a]$
Fact 1.26: orthogonal matrices $M^T M=I$, $\det \pm 1$
Fact 1.27: shear preserves area, distorts shape
Fact 1.28: 2D rotation matrix $[[\cos \theta, -\sin \theta], [\sin \theta, \cos \theta]]$
Fact 1.29: shears don't add linearly in parameters
Fact 1.30: rank = dimension of image (output space)
Fact 1.31: $\det 0 \Rightarrow \text{rank drops} \Rightarrow \text{lower-dimensional image}$
Fact 1.32: Cayley-Hamilton: matrix satisfies own characteristic polynomial
Fact 1.33: diagonalizable powers: $M^n = P D^n P^{-1}$
Fact 1.34: diagonal power = entrywise power
Fact 1.35: $\det(AB) = \det A \cdot \det B$
Fact 1.36: linear algebra = geometry of space transformations

Fact 2.18 isomorphisms non-unique; $V=V$ has infinitely many automorphisms (e.g. $v \mapsto cv$ for any nonzero $c \in \mathbb{F}$); if $\Phi: V \rightarrow V$ is iso and $\Psi: V \rightarrow W$ iso then $\Psi \circ \Phi$ gives infinitely many $V \rightarrow W$ isos; hence existence of one iso implies infinitely many via composition/scaling
Fact 2.24 $\text{span}(S)$ always subspace of V ; closed under addition and scalar multiplication by construction
Fact 2.28/2.30 every finite-dimensional vector space has a basis; basis not unique and in fact any nontrivial finite-dim vector space has infinitely many distinct bases; infinite-dimensional vector spaces also have bases but existence is nontrivial
Fact 2.32 cardinality of any basis of V equals $\dim(V)$; basis size invariant under choice
Fact 2.33 if $U_1, U_2 \leq V$ with $U_1 \cap U_2 = \{0\}$ then $\dim(U_1 \oplus U_2) = \dim(U_1) + \dim(U_2)$ and every $v \in U_1 \oplus U_2$ has unique decomposition $v = u_1 + u_2$
Fact 2.34 every subspace $U \leq V$ (finite-dimensional V) has a complementary subspace $W \leq V$ such that $V = U \oplus W$; complement not unique
Fact basis-isomorphism correspondence any basis z of V defines bijective linear map $z: F^n \rightarrow V$ and conversely any vector space isomorphism $F^n \rightarrow V$ determines a basis via images of canonical basis; thus "basis = coordinate isomorphism"
Fact finite-dim structure $V = F^n$ for unique n ; all finite-dimensional vector spaces are isomorphic to coordinate space of same dimension
Fact uniqueness of zero/inverses in vector spaces: additive identity and additive inverses are unique and scalar zero annihilates all vectors ($0 \cdot v = 0$ vector)

Fact 3.16: $\text{Ker}(T)$ subspace of V
Fact 3.18: T injective iff $\text{Ker}(T) = \{0\}$ iff $\text{Null}(T) = 0$
Fact 3.19: $\text{Rang}(T)$ subspace of W
Fact 3.20: $\text{Rank}(T) = \dim(\text{Rang}(T))$, surjective iff $\text{Rank}(T) = \dim(W)$, $\dim(V) = \text{Null}(T) + \text{Rank}(T)$
Fact 3.21: if $\dim(V) > \dim(W)$ then T not injective
Fact 3.22: for $T: V \rightarrow V$ injective $\Leftrightarrow$ surjective $\Leftrightarrow$ bijective
Fact 3.45: $\{z_i \otimes \zeta_j\}$ forms basis of $L(V, V)$
Fact 3.46: any $T \in L(V, V)$ written $T = \sum_i \lambda_i z_i \otimes \zeta_i$ Fact 3.47: $\dim L(V, V) = \dim(V)^2$
Fact 3.48: components recovered by $\zeta_i(T(z_j)) = T^{\lambda_i} z_j$
Fact 3.52: action of T on v matches matrix multiplication of $T^{\lambda_i} z_j$ with vector components
Fact 3.53: $Mz: L(V, V) \rightarrow \text{Mat}(n \times n)$ is vector space isomorphism
Fact 3.57: $Mz(TS) = Mz(T)Mz(S)$
Fact 3.63: change of basis gives $T_z = P^{-1} \{-1\} T_z P$
Fact 3.65: $Q = y^{-1} z^{-1} \{-1\}$ has matrix P in basis z

Fact 4.1: Operator linear map $T: V \rightarrow V$ on finite-dim V
Fact 4.2: Eigenvector $v \neq 0$ satisfies $Tv = \lambda v$ defining eigenvalue λ unique per v
Fact 4.3: Eigenvectors for same eigenvalue not unique; any scalar multiple also eigenvector; eigenspace is subspace
Fact 4.4: Identity operator has single eigenvalue 1 and every vector is eigenvector
Fact 4.5: Real 2D rotation has no real eigenvectors but over $\mathbb{C}^2$ has eigenvalues $e^{\pm i\theta}$ with complex eigenvectors
Fact 4.6: Real matrices can be viewed as complex-linear maps on $\mathbb{C}^n$ preserving $\mathbb{R}^n$ as subset
Fact 4.7: Eigenvectors with distinct eigenvalues cannot be scalar multiples
Fact 4.8: Eigenvectors with distinct eigenvalues are linearly independent; none is linear combination of others
Fact 4.9: Any set of eigenvectors with distinct eigenvalues is linearly independent
Fact 4.10: In n -dim space any n linearly independent vectors form a basis
Fact 4.11: In n -dim space any $n+1$ vectors are linearly dependent
Fact 4.12: Operator on n -dim space has at most n distinct eigenvalues; if exactly n then eigenvectors form basis
Fact 4.13: Any complex linear operator has at least one eigenvalue and eigenvector
Fact 4.14: Complex polynomials factor completely into linear factors (Fundamental Theorem of Algebra)
Fact 4.15: Existence of eigenvalue via linear dependence of $v, Tv, \dots, T^n v$ leading to polynomial annihilation and non-injective factor
Fact 4.16: Diagonal entries of matrix are M_{ii}
Fact 4.17: Upper-triangular matrix has zero entries below diagonal
Fact 4.18: Every complex operator has a basis making its matrix upper-triangular
Fact 4.19: Same operator under different bases gives similar matrices with same eigenvalues
Fact 4.20: Similar matrices satisfy $A = P^{-1}BP$
Fact 4.21: Similar matrices share eigenvalues and satisfy same polynomial equations
Fact 4.22: Triangularizing basis produces upper-triangular matrix form
Fact 4.23: Every matrix is similar to an upper-triangular matrix
Fact 4.24: Off-diagonal upper-triangular entries cannot in general be eliminated
Fact 4.25: Upper-triangular matrix invertible iff all diagonal entries nonzero
Fact 4.26: Eigenvalues of upper-triangular matrix are exactly its diagonal entries
Fact 4.27: Basis exists where matrix is upper-triangular with eigenvalues on diagonal; trace equals sum of eigenvalues
Fact 4.28: Operator is diagonalizable iff there exists a basis of eigenvectors
Fact 4.29: n distinct eigenvalues imply diagonalizability
Fact 4.30: Repeated eigenvalues may or may not allow diagonalization
Fact 4.31: Example $\text{diag}(1, 2, 2)$ is diagonalizable with eigenspace dimensions 1 and 2
Fact 4.32: Sum of eigenspace dimensions need not equal full dimension (e.g. shear case)
Fact 4.33: Jordan block has λ on diagonal and 1 on superdiagonal
Fact 4.34: Jordan basis gives block-diagonal matrix form
Fact 4.35: Every complex operator has Jordan canonical form unique up to ordering of blocks
Fact 4.36: Jordan canonical form is block-diagonal of Jordan blocks
Fact 4.37: Multiplicity of eigenvalue equals total size of its Jordan blocks (algebraic multiplicity)
Fact 4.38: Characteristic polynomial is product $(x-\lambda)^m$ over eigenvalues with multiplicities
Fact 4.39: Cayley-Hamilton theorem: $\chi_T(T) = 0$
Fact 4.40: In polynomial evaluation of operators, 0 means zero map and 1 identity map
Fact 4.41: Jordan blocks imply nilpotency of $(B-\lambda I)^k$ and give Cayley-Hamilton via annihilation
Fact 4.42: Powers of matrices reduce to combinations of lower powers via characteristic equation
Fact 4.43: Real diagonal JCF corresponds to stretches, reflections, or crushings
Fact 4.44: Real matrices have eigenvalues occurring in complex conjugate pairs
Fact 4.45: Complex diagonal JCF gives stretches, rotations, or crushings; real case combines conjugate pairs into rotations and stretches
Fact 4.46: Real Jordan blocks correspond to stretches combined with generalized shears
Fact 4.47: Any linear operator on $\mathbb{R}^n$ decomposes into stretches, reflections, rotations, crushings, and generalized shears

Fact 5.4: $B(V)$ bilinear forms form vector space under pointwise addition and scalar multiplication
Fact 5.8/5.9: $\zeta_i \otimes \zeta_j$ are bilinear forms and $\{\zeta_i \otimes \zeta_j\}$ forms a basis of $B(V)$
Fact 5.10: any bilinear form $b = \sum_{i,j} \zeta_i \otimes \zeta_j$ with $b_{ij} = b(z_i, z_j)$
Fact 5.18: under basis change with matrix P , bilinear form matrix transforms as $B \rightarrow P^T B P$
Fact 5.29: inner product matrix G is non-singular
Fact 5.33: orthogonal u, v imply $|u+v|^2 = |u|^2 + |v|^2$

Fact 5.35: any u decomposes as $u \parallel v + u \perp v$ with parallel and orthogonal components
Fact 5.36: Cauchy-Schwarz inequality $|g(u, v)| \leq |u||v|$, equality iff linear dependence
Fact 5.37: triangle inequality $|u+v| \leq |u| + |v|$, equality iff same direction with non-negative scalar multiple
Fact 5.40: in $\mathbb{R}^n$, $\text{dot}(u, v) = u^T v$
Fact 5.41: change between orthonormal bases gives orthogonal matrix P with $P^T P = I$
Fact 5.42: orthonormal linear combination has norm squared equal to sum of squares of coefficients
Fact 5.43/44: orthonormal vectors are linearly independent and any n orthonormal vectors in $\dim n$ form a basis
Fact 5.48: Gram-Schmidt change-of-basis matrix is upper triangular
Fact 5.49: every finite-dimensional inner product space has an orthonormal basis
Fact 5.50: Riesz map $\Gamma: V \rightarrow V^*$ is an isomorphism
Fact 5.51: (Riesz representation) every dual vector α equals $g(\cdot, u)$ for unique $u \in V$
Fact 5.53: operator T corresponds canonically to bilinear form $g(u, v) = g(u, Tv)$ (bijection in real case)
Fact 5.54: in an orthonormal basis, matrix of operator equals matrix of corresponding bilinear form
Fact 5.56: dot product on $\mathbb{C}^n$ $u \cdot v = u^H v$, $u \cdot u = aa^H + bb^H + cc^H$ real ≥ 0 , length via $\sqrt{g(v, v)}$
Fact 5.57: sesquilinear form s linear in first slot $s(cu, v) = cs(u, v)$, conjugate-linear in second $s(u, cv) = \bar{c}s(u, v)$
Fact 5.58: matrix of sesquilinear form in basis z_i is $s_{ij} = s(z_i, z_j)$
Fact 5.59: change of basis for sesquilinear form matrix $M \rightarrow P^H M P$
Fact 5.60: complex inner product space (V, g) sesquilinear g positive definite, $g(u, v) = \overline{g(v, u)}$ conjugate symmetry, $g(u, u) \geq 0$ equality iff $u=0$
Fact 5.61: unitary matrix = change between orthonormal bases, columns orthonormal, satisfies $P^H P = I$
Fact 5.63: Riesz correspondence gives bijection between linear operators and sesquilinear forms
Fact 5.64: in orthonormal basis matrix of Riesz form is complex conjugate of operator matrix
Fact 5.65: Schur triangularization: any operator on finite-dim complex inner product space has orthonormal basis with upper-triangular matrix, eigenvalues = diagonal
Fact 5.66: Schur for sesquilinear forms: exists orthonormal basis where matrix is upper-triangular
Fact 5.67: any complex matrix $M = P^H M P$ form with unitary P and upper-triangular U (unitary triangularization)
Fact 5.68: Hermitian sesquilinear form $s(u, v) = \overline{s(v, u)}$, antiHermitian $s(u, v) = -\overline{s(v, u)}$
Fact 5.69: Hermitian matrix $M^H = M$, antiHermitian $M^H = -M$
Fact 5.70: spectral theorem Hermitian operator $\Rightarrow$ orthonormal eigenbasis, all eigenvalues real
Fact 5.71: Hermitian matrices diagonalizable by unitary P , D real diagonal
Fact 5.73: real symmetric operators have orthonormal eigenbasis with real eigenvectors
Fact 5.74: symmetric matrices diagonalizable by orthogonal O , $D = O^H M O$ real diagonal
Fact 5.76: spectral decomposition $S = \sum \lambda_i z_i \otimes \zeta_i$ for symmetric bilinear form
Fact 5.77: symmetric bilinear form positive definite $\Leftrightarrow$ all eigenvalues > 0
Fact 5.78: Hessian test: zero gradient + positive eigenvalues of Hessian $\Rightarrow$ local minimum
Fact 5.79: Clairaut-Schwarz theorem: mixed partial derivatives commute $\Rightarrow$ Hessian symmetric
Fact 5.80: any sesquilinear form $s = \sum_{i,j} \zeta_i \otimes \zeta_j$ in basis-dual basis form
Fact 5.81: any complex matrix corresponds to a sesquilinear form in some basis
Fact 5.82: Hermitian spectral decomposition $H = \sum \lambda_i z_i \otimes \zeta_i$ with real λ_i

Fact 6.1: Multilinear form degree m is a map $V^m \rightarrow \mathbb{R}$ linear in each slot
Fact 6.2: Two-form is a bilinear form satisfying $\Psi(u, v) = -\Psi(v, u)$
Fact 6.3: Wedge product $\alpha \wedge \beta = \alpha \otimes \beta - \beta \otimes \alpha$ and $\alpha \wedge \alpha = 0$
Fact 6.4: Any 2-form $\Psi = 1/2 \sum_{i,j} \zeta_i \wedge \zeta_j$ with $\psi_{ij} = \Psi(z_i, z_j)$ and $\psi_{ij} = -\psi_{ji}$
Fact 6.5: Dimension of space of m -forms on n -dim space is $C(n, m)$
Fact 6.6: Space of n -forms on n -dim space is 1-dimensional
Fact 6.7: Define $\Delta(T)$ by ${}^T \Delta = \Delta(T) Q$ for any n -form Q
Fact 6.8: $\Delta(T)$ equals product of eigenvalues of T with multiplicity
Fact 6.9: $\Delta(T)$ invariant under basis change and equals $\Delta(T^*)$ for any matrix rep
Fact 6.11: $\Delta(TS) = \Delta(T)\Delta(S)$ Fact 6.12: $\Delta(T) \neq 0$ iff T is bijective
Fact 6.13: $\Delta(T-\lambda I) = 0$ for every eigenvalue λ Fact 6.15: Determinant defined as $\Delta(T)$
Fact 6.16: Volume form $\Theta = |\omega z|$ where $\omega z = \zeta_1 \wedge \zeta_2 \wedge \dots \wedge \zeta_n$
Fact 6.17: Orientation given by sign of ωz ; ωz changes by ± 1 under basis change
Fact 6.18: Linear transformation scales volume by $|\Delta(T)|$
Fact 6.19: $\Theta = \sqrt{|\Delta(G)|} \eta_1 \wedge \eta_2 \wedge \dots \wedge \eta_n$ for Gram matrix G
Fact 6.20: Bilinear form determinants change under basis via PTP while operator determinants are invariant under similarity